Burning it all

16a

When substances burn or combust, they react with oxygen to form *oxides*. Usually oxygen from the air is used, but at times oxygen is supplied from special containers. Extra amounts of oxygen ensure that all substances are completely oxidised.

1 Coal is essentially made of carbon. When carbon reacts with oxygen gas from the air, it produces carbon dioxide. Here is the reaction equation in symbols: $C + O_2 \rightarrow CO_2$
Write the reaction equation in words.

2 During a science experiment, you held a silver piece of magnesium ribbon into a burner flame, and it suddenly lit up giving off a very bright white light. When the reaction was finished, you were left with a white powder. The magnesium had reacted with oxygen gas.

a Write the word equation and symbol equation for that reaction.

b When you count the number of oxygen atoms in the reaction above, you should find two oxygen atoms on the left side (O_2 is made up of two O atoms) but only one on the right side (MgO has only one oxygen atom). That means that according to your reaction equation you have lost one oxygen atom during this reaction. However, that is not possible. In order to balance reaction equations, chemists write large numbers in front of the substances until all atoms on the left side of the equation are also on the right side. Let's do this with your equation.

$Mg + O_2 \rightarrow MgO$	*left side*: one Mg atom, two O atoms; *right side*: one Mg atom, one O atom – not balanced
$Mg + O_2 \rightarrow 2\ MgO$	*left side*: one Mg atom, two O atoms; *right side*: two Mg atoms, two O atoms – not balanced
$2\ Mg + O_2 \rightarrow 2\ MgO$	*left side*: two Mg atoms, two O atoms; *right side*: two Mg atoms, two O atoms – balanced

Write the balanced symbol equation for the burning of magnesium.

3 The gas that comes out of the gas taps in the laboratory is mainly methane gas, a hydrocarbon.

a State the two elements present in hydrocarbons.

b When methane burns in oxygen, both elements react with oxygen to produce two oxides. Translate the following symbol equation into a word equation:

$CH_4 + O_2 \rightarrow CO_2 + H_2O$

c Balance the symbol equation by writing large numbers in front of some of the substances in the equation. When the equation is balanced, there should be four oxygen atoms and four hydrogen atoms on both sides of the equation.

$CH_4 + \ O_2 \rightarrow \ CO_2 + \ H_2O$

4 When the valve of the Bunsen burner is closed, methane gas mixes with air at the top of the barrel. However, there is not enough oxygen available to oxidise all carbon. That explains the black smudge that appears on the test tube when you heat it with the yellow safety flame.

a Write a word equation and a symbol equation for the incomplete combustion of methane gas.

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b Write the balanced symbol equation.

c When the valve of the Bunsen burner is opened, complete combustion of methane occurs, ie both elements present in methane are fully oxidised.
Write the word equation for the complete combustion of methane gas.

5 Petrol is also a hydrocarbon. However, it contains 8 carbon atoms and 18 hydrogen atoms and it is called *octane*. When a car engine is working properly, all carbon and hydrogen is oxidised.

a Write the word equation and symbol equation for the complete combustion of octane.

b Write the balanced symbol equation.

c When a car engine is not working properly, poisonous carbon monoxide gas and black carbon powder, as well as carbon dioxide and water, are produced.
Write the word equation and symbol equation for the incomplete combustion of petrol.

d Write the balanced symbol equation for the incomplete combustion of octane. (Hint: Balance the carbon first, then the hydrogen.)

e Suggest how the complete combustion of petrol could be ensured in an engine.

6 White table sugar is a carbohydrate called sucrose. Sucrose is made up of many glucose molecules. The formula of glucose is $C_6H_{12}O_6$.

a Name the elements present in a carbohydrate.

b The cells in your body receive energy during the process called respiration. In respiration glucose reacts with oxygen gas.
Write the word equation and a symbol equation for respiration.

c Write a balanced symbol equation for respiration.

d Explain how glucose and oxygen are transported to muscle cells in your body.

e Explain how your body gets rid of the two waste products of respiration.

f When your body has to work very hard, for example during a strenuous sports event, there is not enough oxygen available for the complete oxidation of glucose. As a result, the two waste products are not produced and taken away. Instead other chemicals are produced which stay in your muscles and cause muscular pain. The main chemical which gives you sore muscles is called lactic acid. Suggest how you could prevent or ease sore muscles.

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